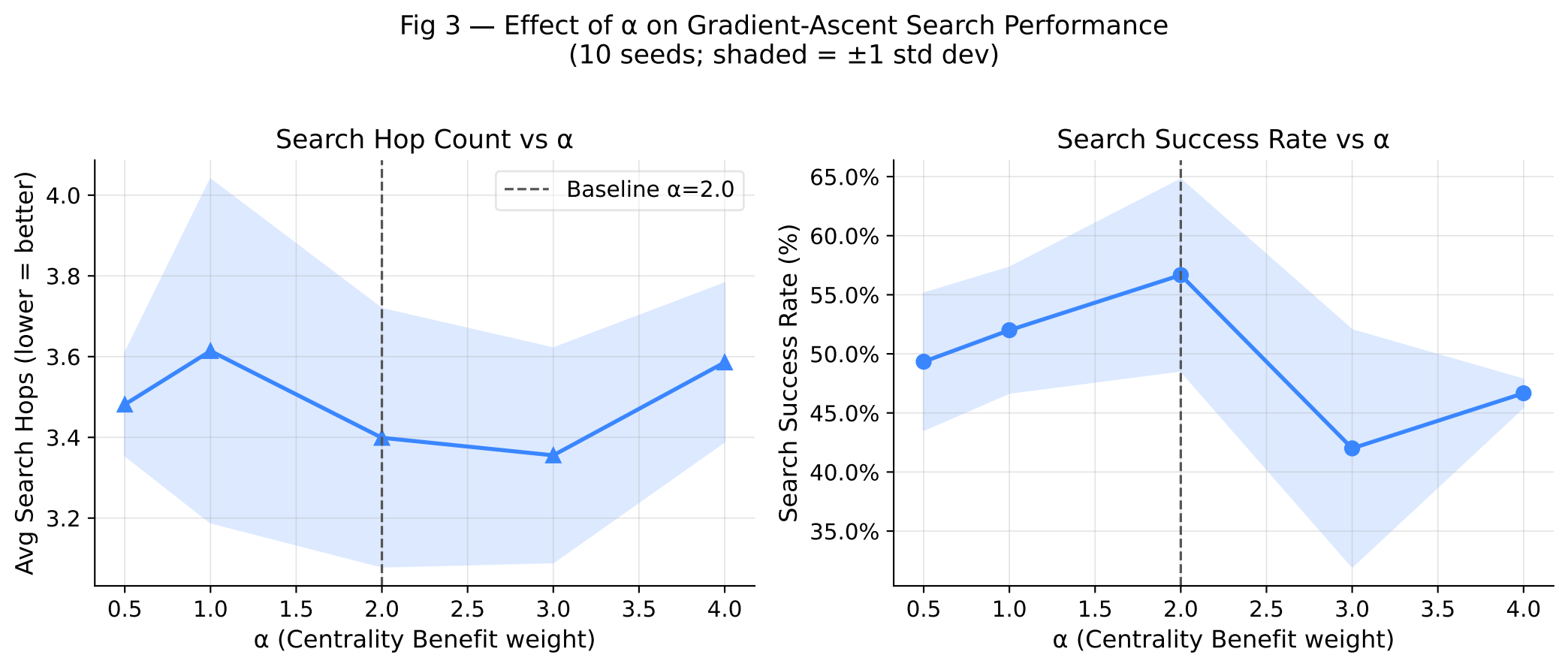
Fig 3 — Effect of α on Gradient-Ascent Search Performance
(10 seeds; shaded = ±1 std dev)
Search Hop Count vs α
Avg Search Hops (lower = better)
Baseline α=2.0
4.0
3.8
3.6
3.4
3.2
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
α (Centrality Benefit weight)
Search Success Rate vs α
Search Success Rate (%)
65.0%
60.0%
55.0%
50.0%
45.0%
40.0%
35.0%
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
α (Centrality Benefit weight)